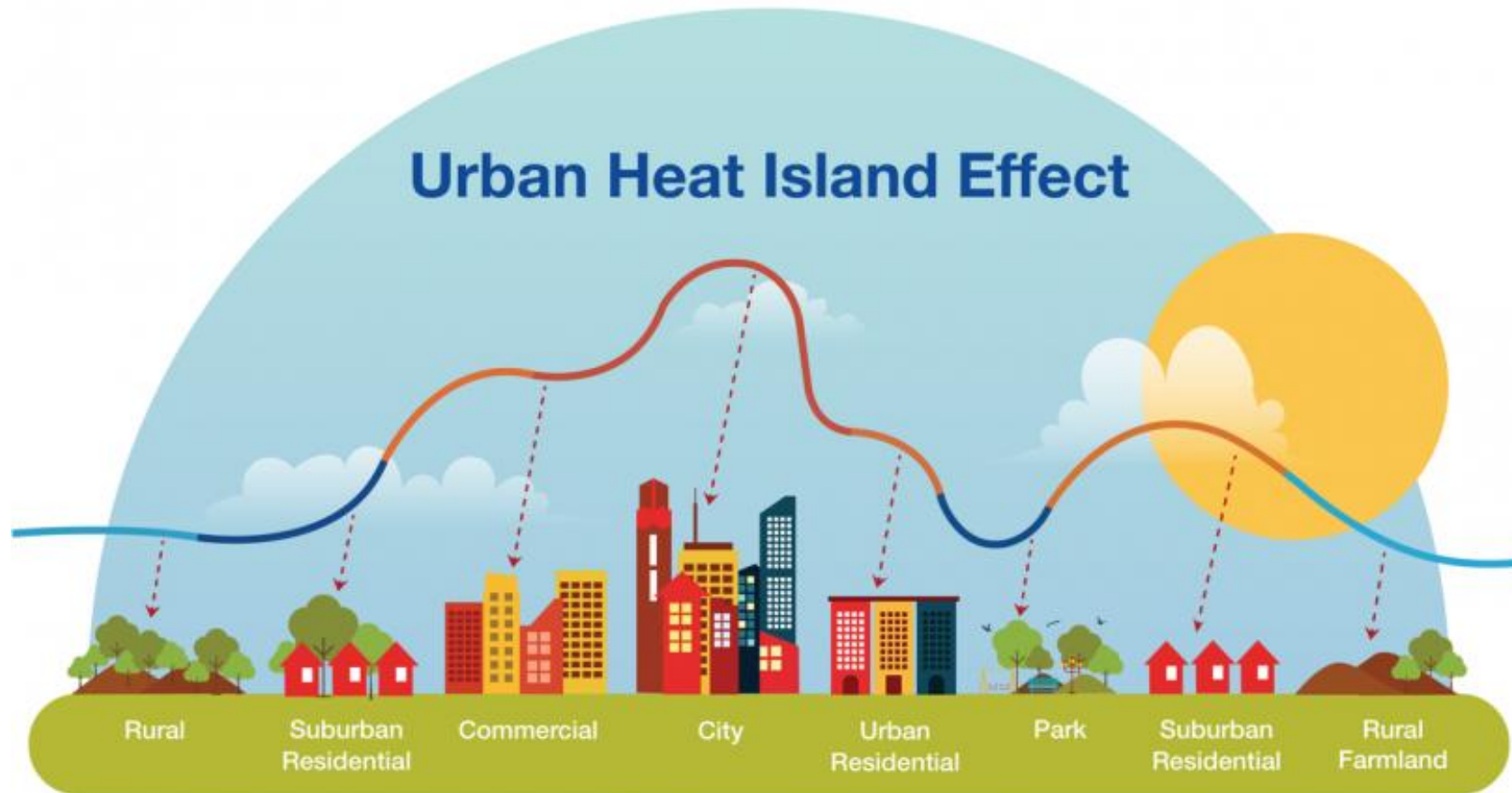


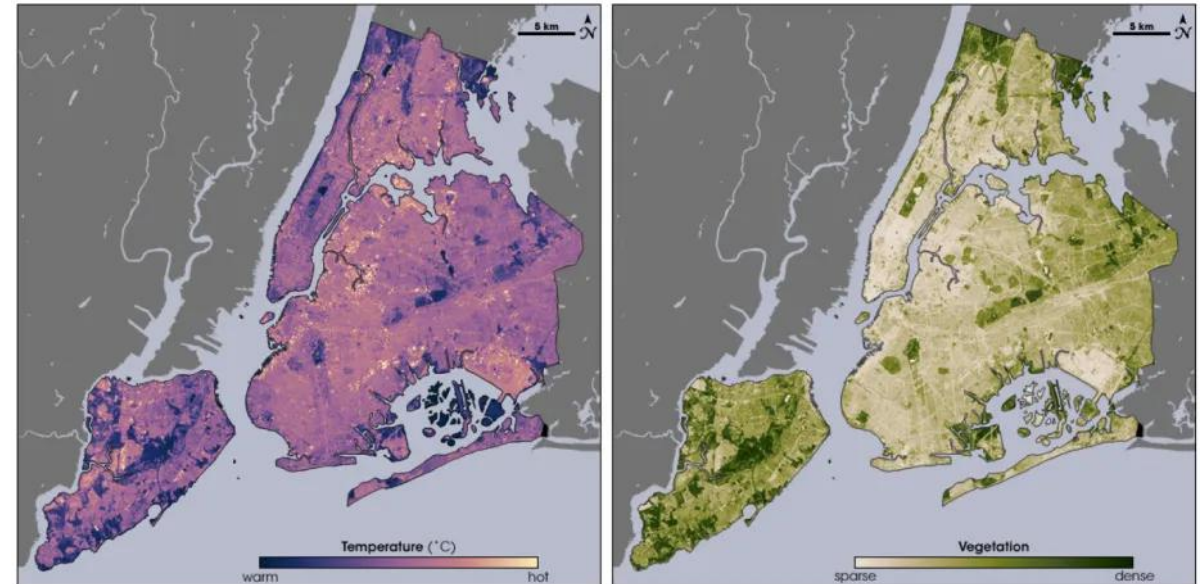
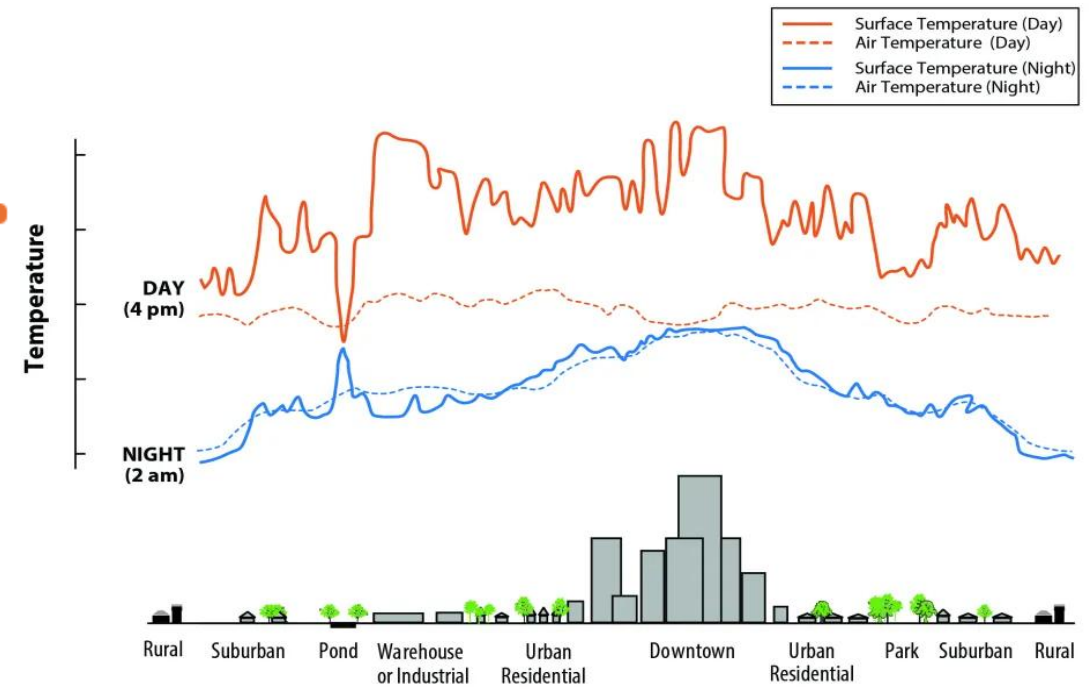
Understanding Urban Heat Through the Modeling of Temperature and Population



Authors: Nikita Neeli, Harsh Gupta, Thamim Parvin, Deborah Ugaldes, and Vishaka Venkatraman

Problem Statement

- **Central Phenomenon:**
 - The urban heat island effect, where industrial cities experience higher temperatures than the surrounding rural areas. See Figure 1
- **Study Objective:**
 - Explore the impact of population size on temperature fluctuations in cities over time.
- **Methodology:**
 - Build predictive models that define temperature patterns using demographic and spatial information.
- **Expected Impact:**
 - To provide planning insights for future urban development and environmental management to limit warming.



Dataset Collection and Description

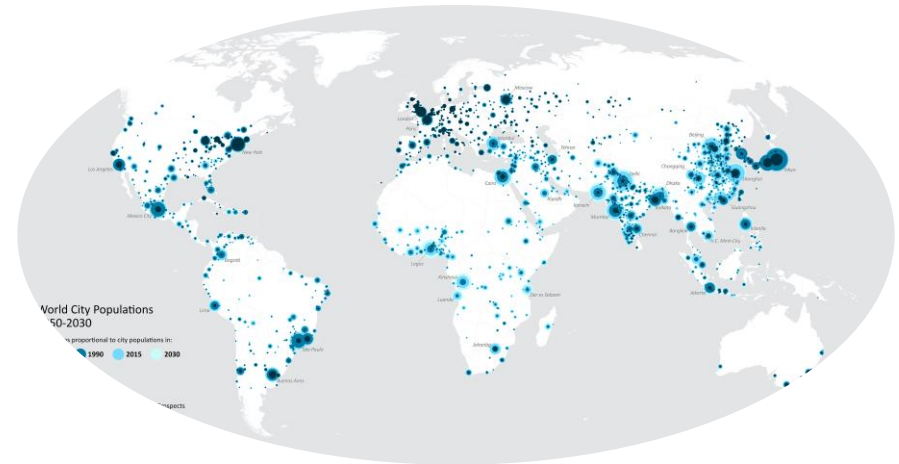
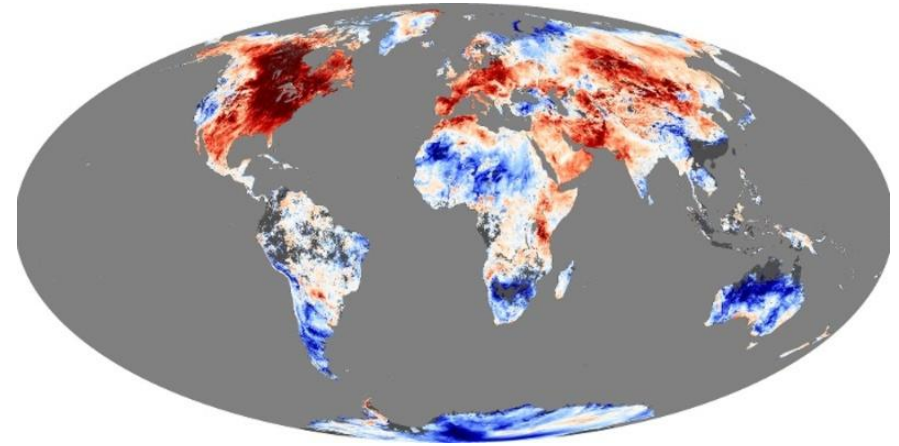
- **Dataset 1: "Global Land Temperatures by Major Cities"**
 - It contains monthly average temperature records for cities around the world.
 - Key variables: Average temperature (dependent variable), date, city, latitude and longitude.
 - See Figure 2.
- **Dataset 2: "World Cities Population Database"**
 - It contains city populations and country identifiers.
 - Used to determine the level of urbanization.
 - See Figure 3

Data Combination:

Both datasets were combined using common city and country identifiers.

Merged dataset size: 228,175 monthly records from 100 cities

Year range: 1743 to 2013

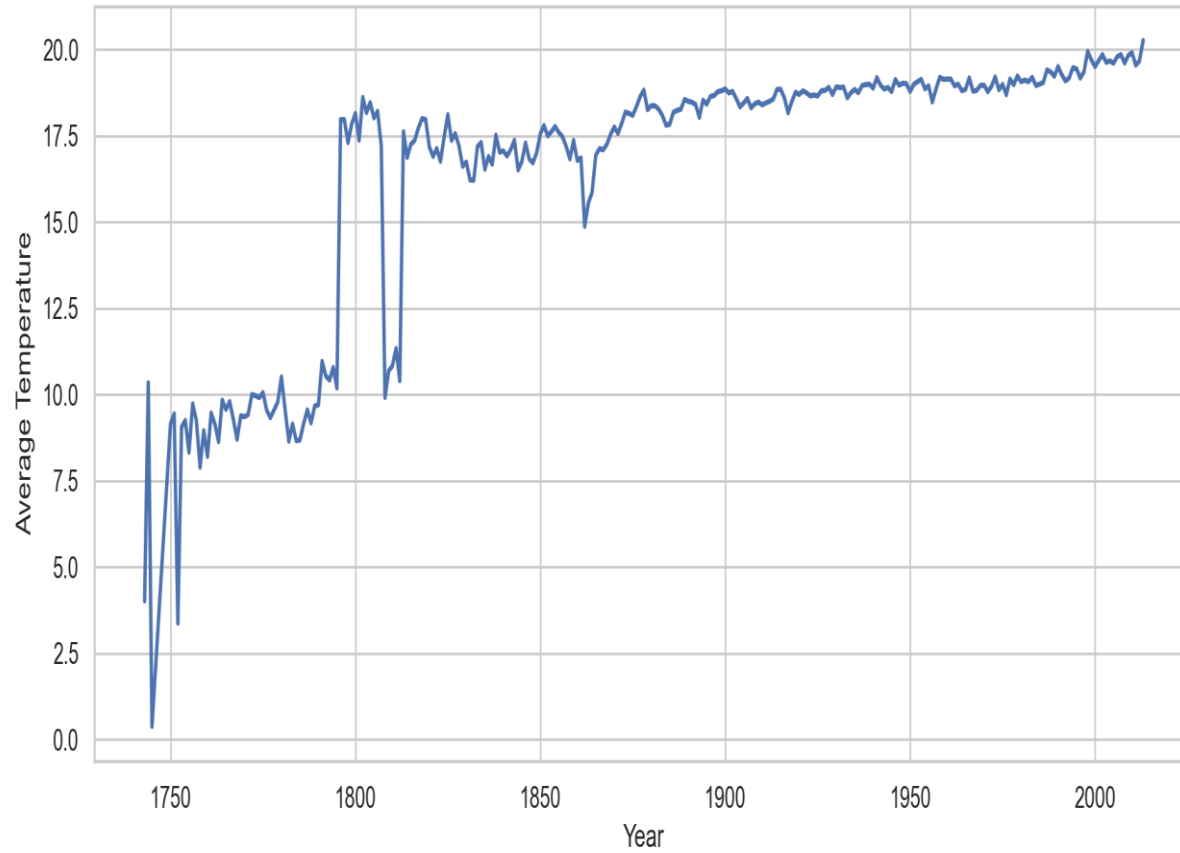


Applied Data Preprocessing Techniques

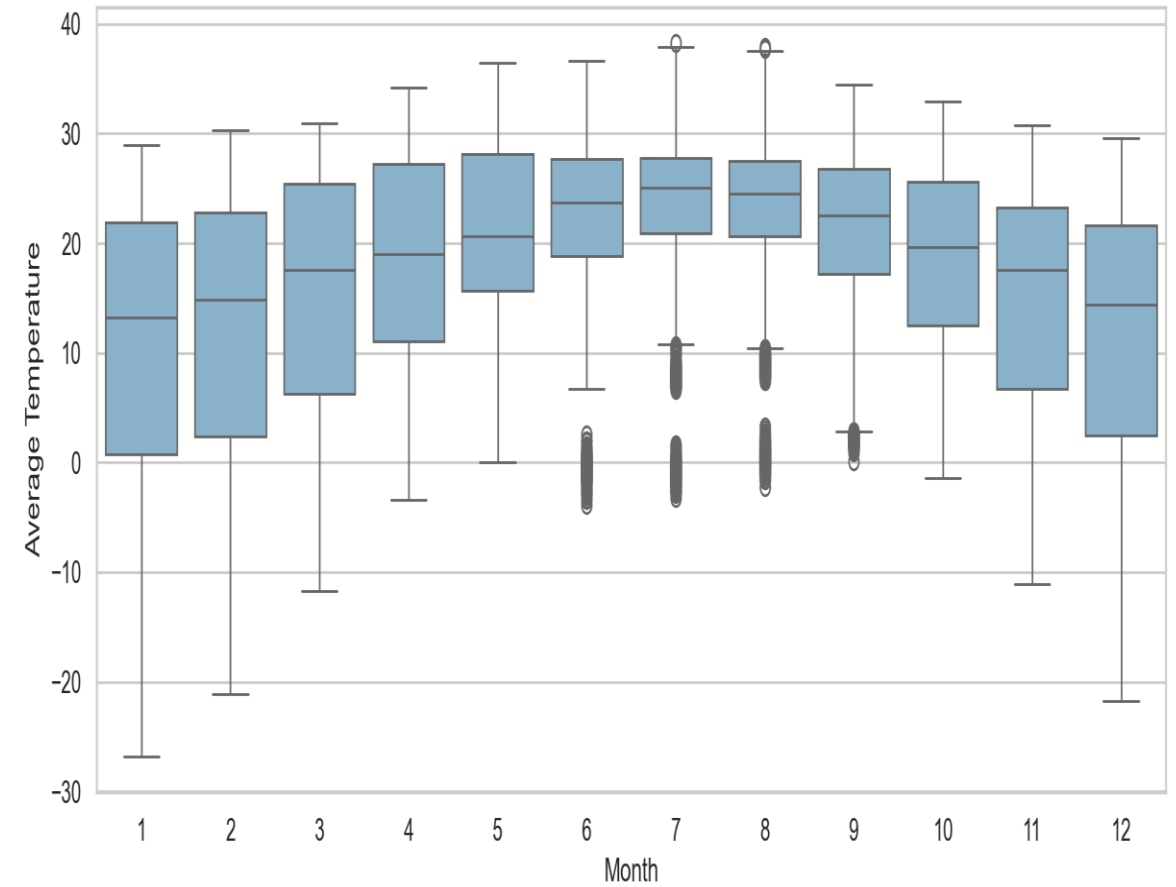
- **Missing Values:** Handling missing temperature observations through elimination of the observation (rows)
- **Temporal Variables:** Converting the date column to a numeric format (extracting year and month) for time-based analysis.
- **Categorical/Numeric Variables:**
 - Reformatting the population column (removal of commas).
 - Transformation of Longitude and Latitude values ("40.7N") to numerical values and cardinal direction.
- **Preparation for the Fusion:** Standardization of key variables (uniform capitalization, removal of unusual characters) for proper merging by city and country.

Visual Examination (EDA)

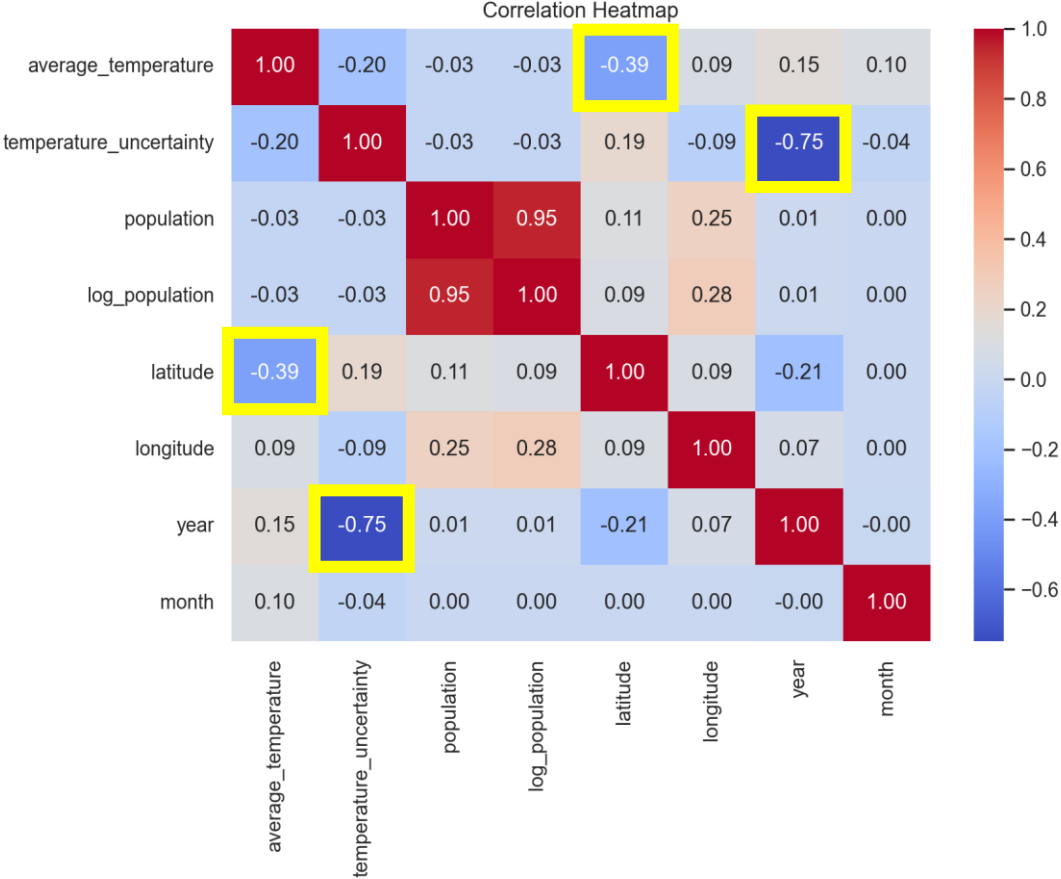
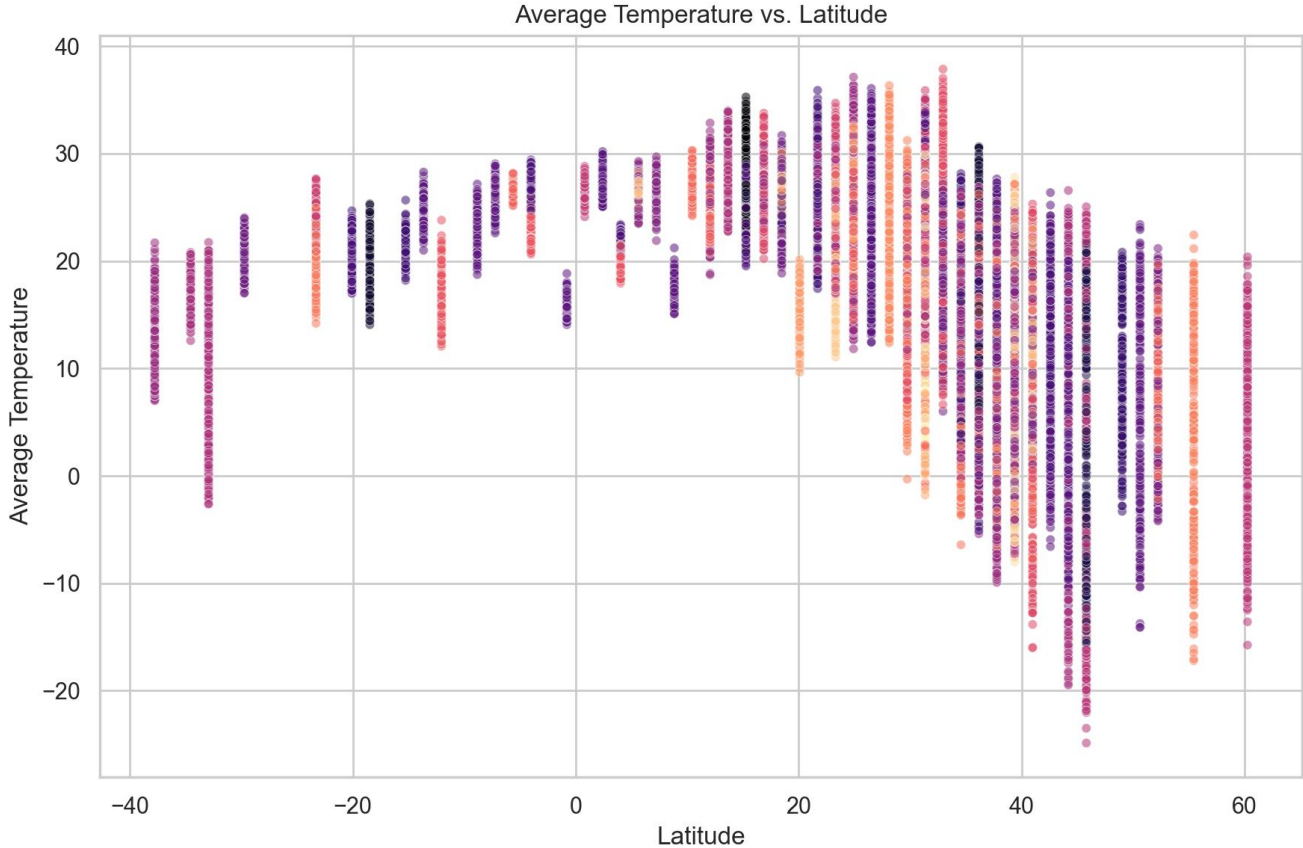
Mean Major-City Temperature Over Time



Monthly Temperature Distribution



Visual Examination (EDA)



- **Reference Models:** Creation of null models to predict average temperatures and establish a baseline.
- **Simple Models:** Multiple linear regression to capture relationships between key features such as population and latitude.
- **Models of Intermediate Complexity:** Polynomial regression or decision trees to model relationships that are non-linear.
- **Advanced Model:** Implementation of neural networks to capture complex synergies between variables.
- **Why we chose the models:**
 - Linear Regression: Initially chosen for its simplicity and interpretability.
 - Flexible Models: Polynomial regression and neural networks are used to capture nonlinear patterns, especially given the temporal and spatial nature of the data.
 - Baseline (Null Model): It provides an essential benchmark for assessing the gain in predictive power by incorporating urban factors such as population.

Dataset Overview

- **Records:** 228,175
- **Cities:** 100
- **Countries:** 49
- **Years Covered:** 1743 to 2013
- **Population Range Used For Matching:** 1,200,000 to 22,315,474
- **Mean average temperature:** 18.13

Five Warmest Cities by Mean Temperature

city	country	avg_temperat	population
Umm Durmar	Sudan	29.08	1,200,000
Madras	India	28.42	4,681,087
Jiddah	Saudi Arabia	27.69	2,867,446
Ho Chi Minh	Vietnam	27.19	8,993,082
Bangkok	Thailand	27.16	5,104,476

Five Coolest Cities by Mean Temperature

city	country	avg_temperat	population
Harbin	China	3.63	5,242,897
Saint Petersburg	Russia	3.92	5,351,935
Moscow	Russia	4.00	10,381,222
Montreal	Canada	4.45	1,762,949
Changchun	China	4.92	4,714,996

Feature Selection

Subset search used only the training period with time-series cross-validation and linear-regression RMSE.

Selected Features:

- log_population, latitude, longitude, abs_latitude, year, month_sin, month_cos

Candidate Feature Flags

feature	selected_for_final
log_population	True
latitude	True
longitude	True
abs_latitude	True
year	True
month_sin	True
month_cos	True

Top CV Subsets

rank	feature_count	cv_rmse_mean	cv_rmse_std
1	7	6.51	0.10
2	6	6.51	0.10
3	6	6.52	0.06
4	6	6.52	0.09
5	5	6.52	0.06

Winning Subset:

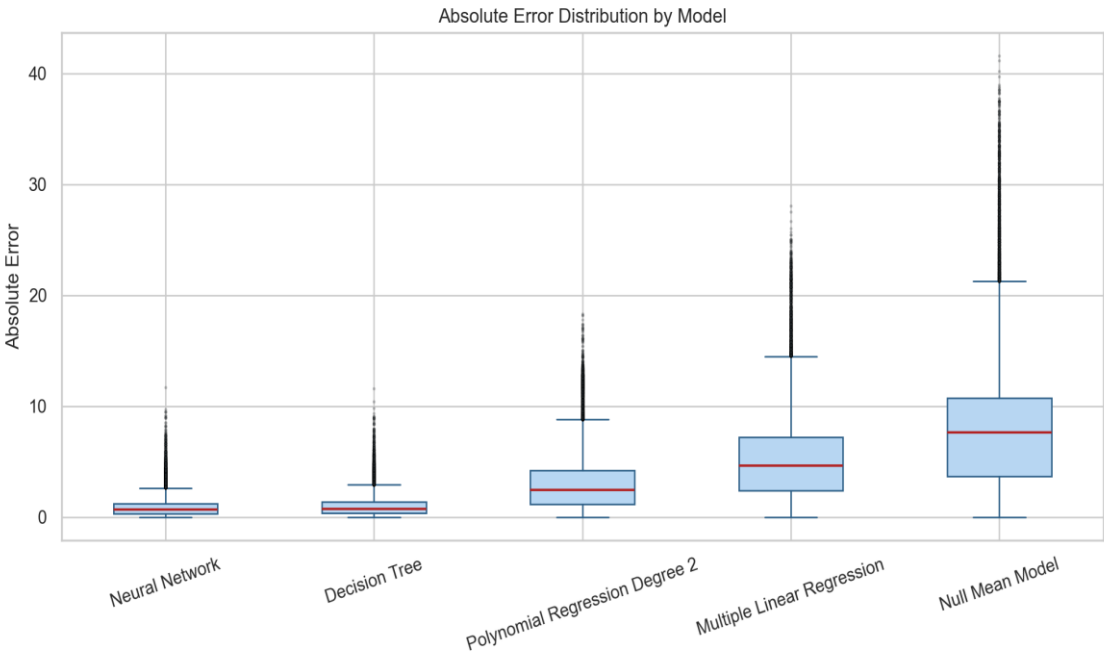
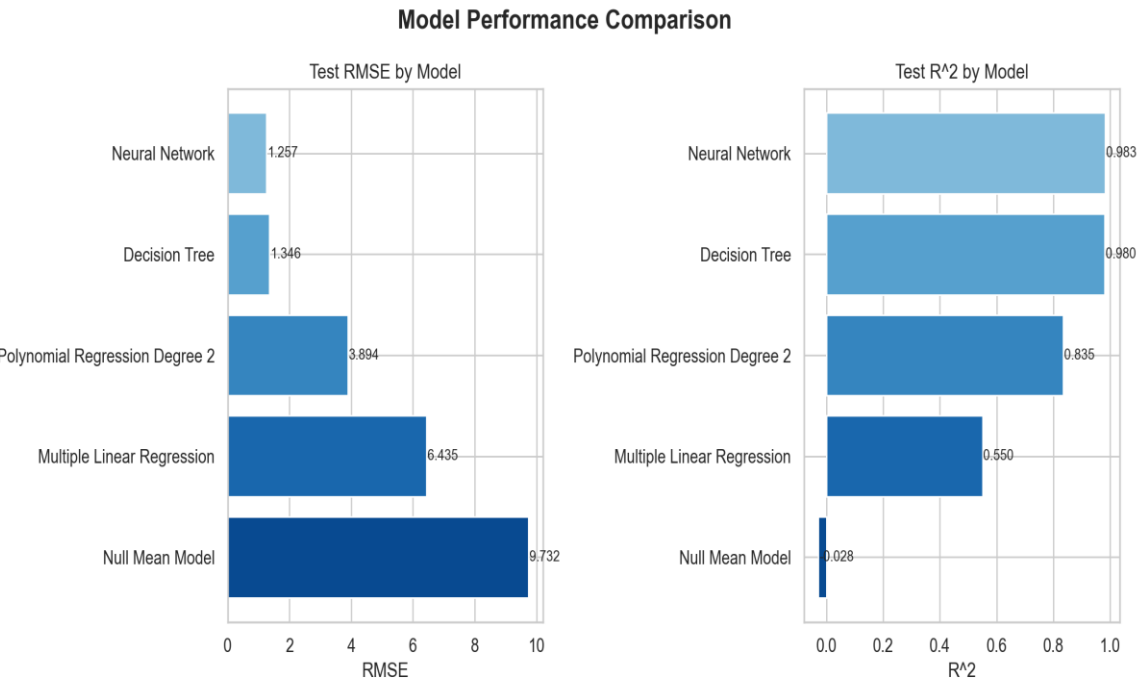
- log_population, latitude, longitude, abs_latitude, year, month_sin, month_cos

Feature choice is now based on train-only evidence rather than intuition alone, which makes the workflow easier to defend.

Results Report

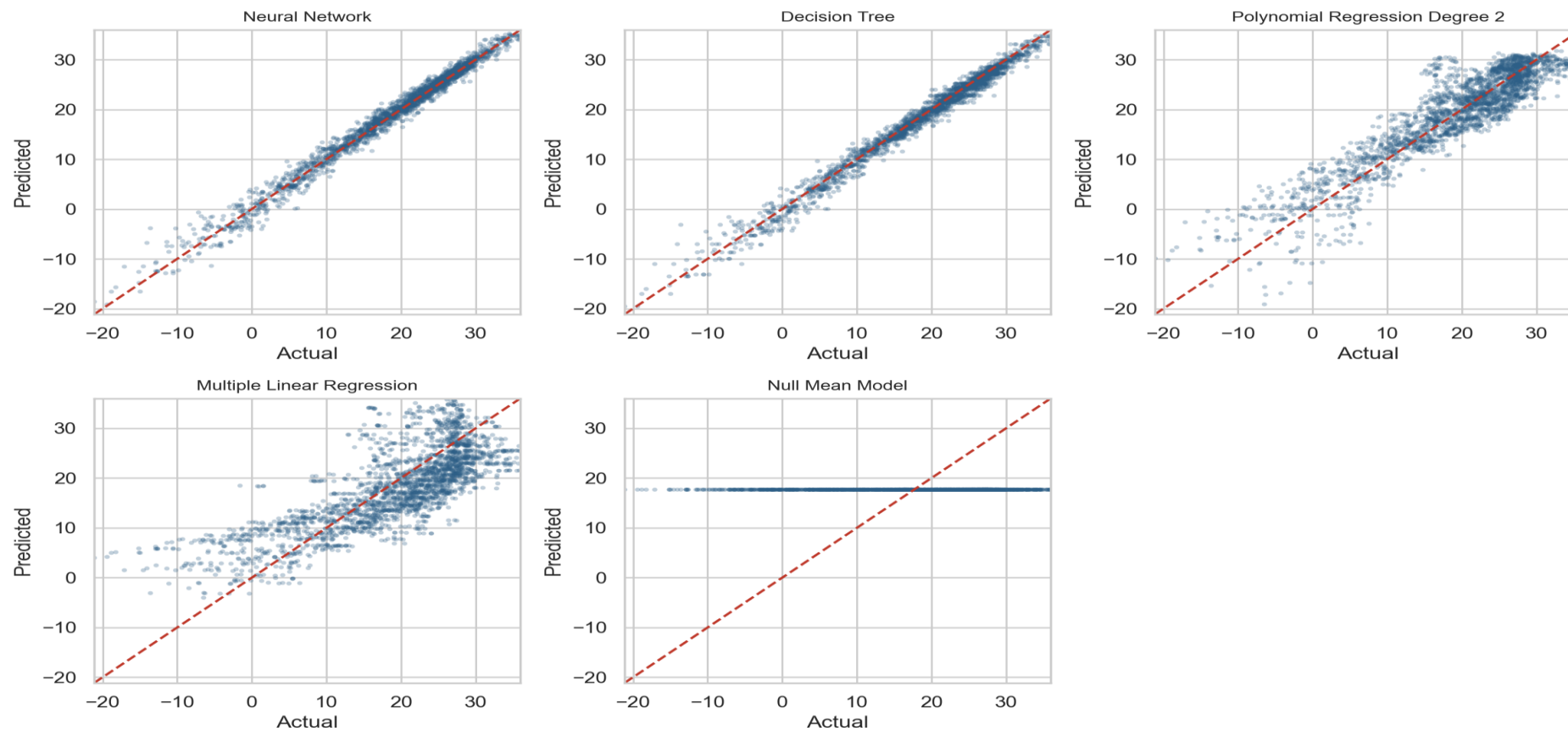
Five predictive models were evaluated. Their efficacy was measured using a chronological train-test split, training on data from 1743 to 1959 and testing on the 1960–2013 period.

Model	RMSE	R^2
Neural Network	1.257	0.983
Decision Tree	1.346	0.980
Polynomial Regression Degree 2	3.894	0.835
Multiple Linear Regression	6.435	0.550
Null Mean Model	9.732	-0.028

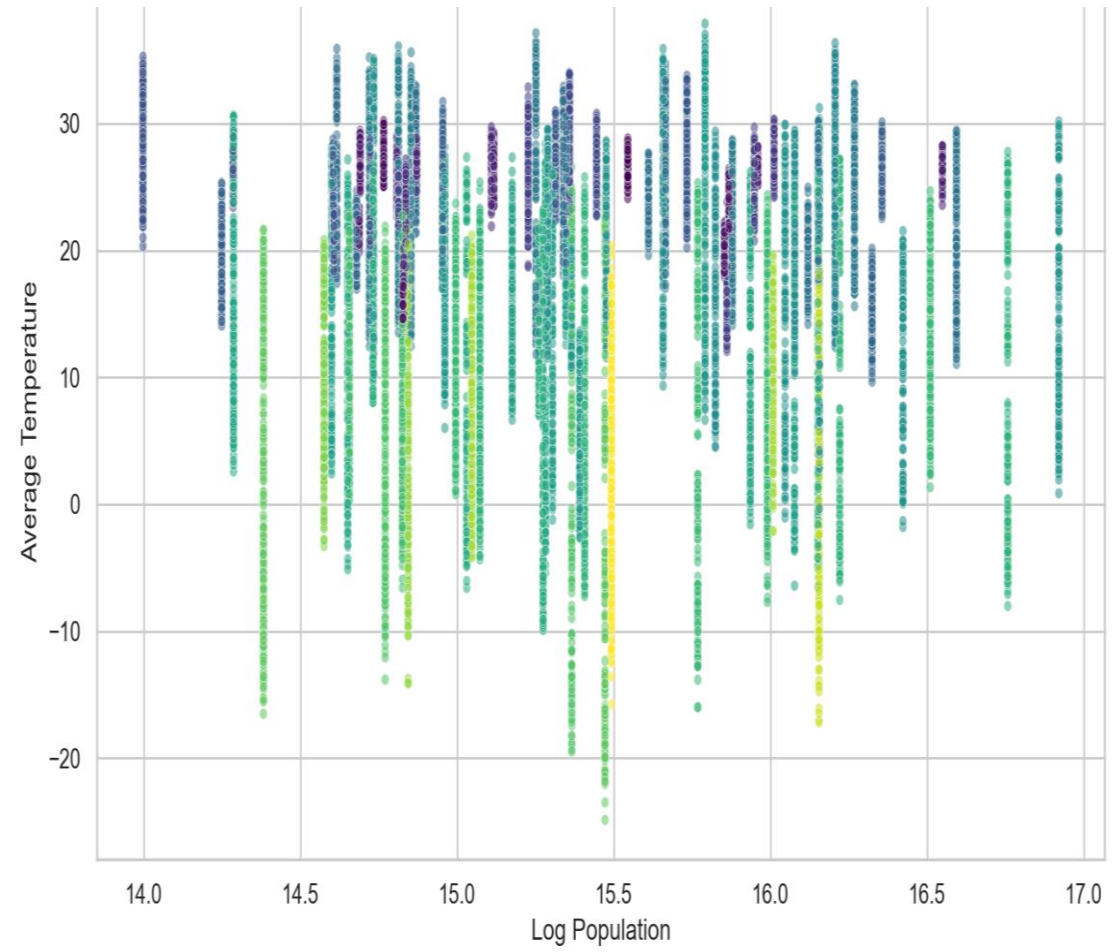
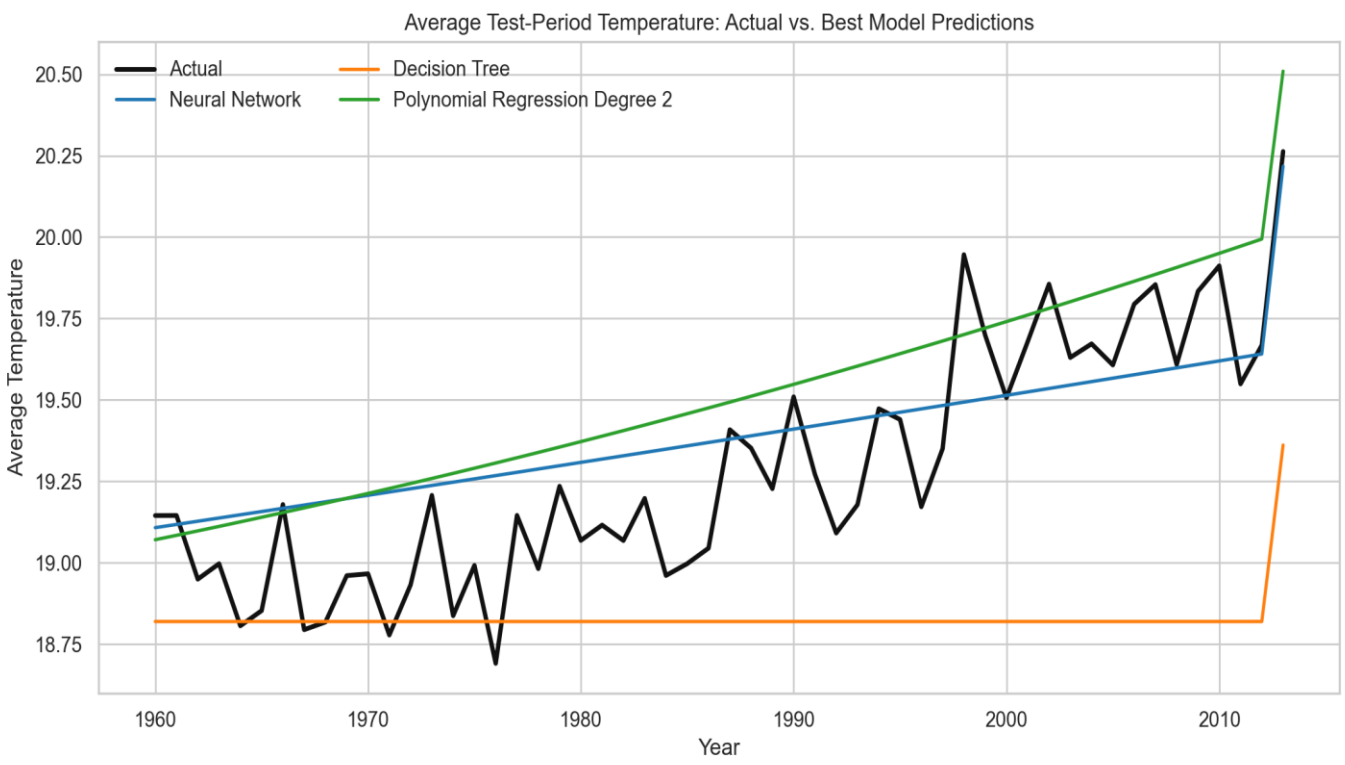


Model Comparison Visualization

Actual vs. Predicted Temperature by Model



Model Comparison Visualization



Interpretation of Results

- **Discussion**

- The strongest overall model was neural network, showing that temperature is predictable from the chosen features.
- Latitude has a much stronger raw relationship with temperature (-0.393) than log population (-0.030).
- Year also has a positive association with temperature (0.147), which is consistent with a warming trend over time.

- **Conclusions**

- The population dataset is modern and effectively static, so it does not track historical city growth year by year.
- That means population works better here as a proxy for city scale than as a direct causal explanation of long-run temperature change.
- The project supports a predictive story more strongly than a causal urban-heat claim.

Study Recommendations



- **Urban Planning Recommendations:**

- Strategies to mitigate the increase in temperatures in densely populated areas.
- Population distribution management.
- Increase green spaces or improve infrastructure to reduce heat retention.

- **Future Research:**

- Incorporate additional environmental variables (vegetation cover, land use, pollution levels) to improve accuracy.
- Expand the dataset to include more cities or longer time periods to strengthen the conclusions.